

CLIMATE CHANGE

WILL IT BE OUR END?

CLIMATE CHANGE

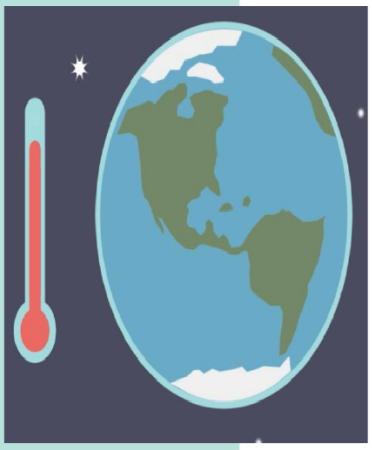
TOPICS

- WHAT IT **MEANS**
- **GLOBAL WARMING** AND ITS **CAUSES**
- CLIMATE CHANGE & ITS CAUSES
- **IMPACT** ON THE OVERALL ENVIRONMENT
- **MEASURES** TO BE TAKEN



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CLIMATE CHANGE



CLIMATE

Weather: The overall condition of a place over a short period of time, considering factors like temperature, rainfall, etc.

Climate: Average of the weather of a particular region

- Measured over a long time, for example, 30 years
- A huge area is taken into consideration; it can be a country, a set of countries, or an entire continent
- Doesn't change frequently

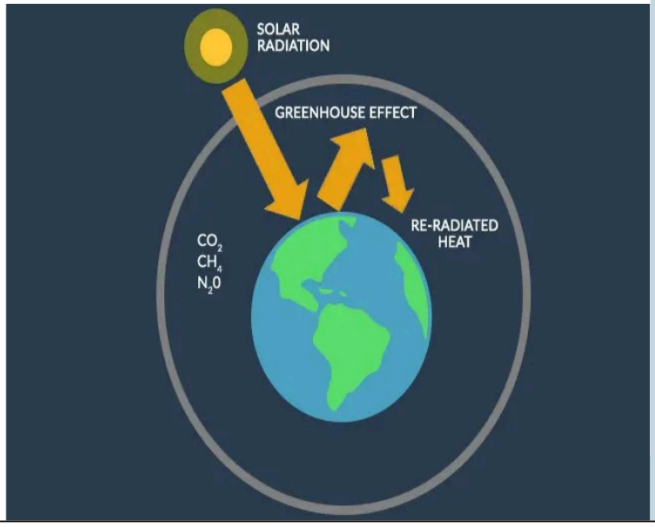
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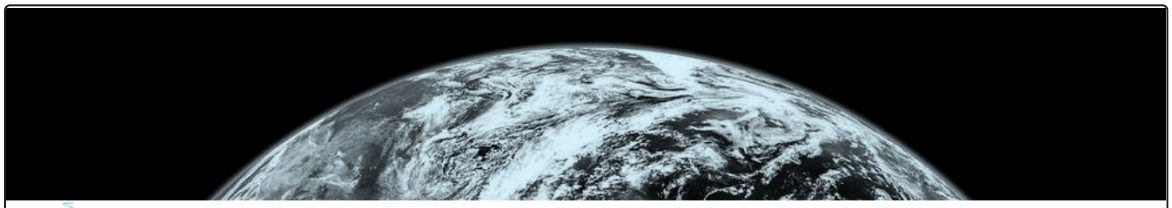
GREENHOUSE EFFECT AND CLIMATE CHANGE

THE GREENHOUSE EFFECT IS THE WAY IN WHICH HEAT IS TRAPPED CLOSE TO THE EARTH'S SURFACE BY GREENHOUSE GASES.

GHGS INCLUDE CO2, CH4, NO2, WATER VAPOR ETC.



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CLIM

STEP 1

During the day, solar radiation reaches the Earth's atmosphere and a portion of it is reflected to space

STEP 2

Some of the heat is absorbed by the land and oceans

STEP 3

At night, starts radiating heat that goes towards the upper atmosphere but is trapped by gases

STEP 4

This excessive heat trapping causes the Earth's temperature to rise over time

STEP 5

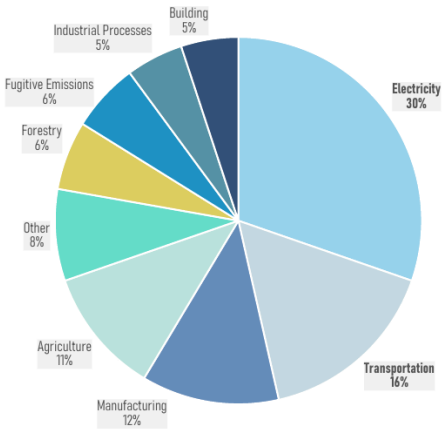
Human activities like burning fossil fuels increase GHGs and the greenhouse effect continues ever so rapidly

HOW IT HAPPENS

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CLIMATE CHANGE

GHG EMISSIONS BY SECTOR



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PRESENTATION TITLE

Kyoto Protocol

The Protocol was adopted by Parties to the UNFCCC in 1997 and entered into force in 2005.

The Protocol "recognizes that developed countries are principally responsible for the current high levels of GHG emissions in the atmosphere because of more than 150 years of industrial activity and places a heavier burden on developed nations under the principle of 'common but differentiated responsibilities "

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PRESENTATION TITLE

Carbon Footprint

- The term "**carbon footprint**" is often used as shorthand for **carbon (usually in tones) being emitted from organization.**

The carbon footprint



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PRESENTATION TITLE

CARBON TRADING

Carbon Trading is a scheme where firms (or countries) buy and sell carbon permits as part of a program to reduce carbon emissions. Usually, firms are given a certain quota to pollute a certain amount. If they wish to pollute more than their allowance, then they must buy more permits. If they pollute less than their quota, they can sell their spare permits on the market. Thus, there is an incentive to reduce pollution and find the most efficient way of dealing with pollution.

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CLIMATE CHANGE

THE SEVEN (7) THEORIES

1. **Anthropogenic Global Warming(AGW) theory:** This theory of climate change contends that human emissions of greenhouse gases, principally carbon dioxide (CO2, methane, and nitrous oxide, are causing a catastrophic rise in global temperatures. The mechanism whereby this happens is called **the enhanced greenhouse effect**. It includes the activities of human beings. For example,
   - Usage of Non-renewable energy resources
   - Usage of artificial fertilizers and insecticides
   - General environmental pollution
2. **Human Forcing Besides Greenhouse Gases:** Human activities that directly affect the Earth's surface, like building cities and deforestation, coastal developments, jet contrails, etc., for transformation of the Earth's surface.

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PRESENTATION TITLE

- **3. Bio-thermostat theory:** Rising temperatures and CO2 levels trigger biological and chemical responses in the atmosphere that have a cooling effect. The more CO2 there is in the air, the better plants grow, and the more CO2 they remove from the air and store in their leaves, branches, trunks, and roots, as well as in the soil beneath the plants – a suite of processes called "sequestration." Higher temperatures also increase carbon sequestration rates: Carbonyl sulphide (COS), diffuse light, iodo compounds, dimethyl sulphide, and other aerosols.
- **4. Cloud Formation & Albedo:** postulates that changes in the formation and albedo of clouds create negative feedbacks that cancel out all or nearly all the warming effect of higher levels of CO2.How the solar reflectivity of snow has an impact on climate change. **Albedo** is a fraction of light reflected by a body or surface. It is commonly used in astronomy to describe the reflective properties of planets, satellites, and asteroids. Albedo = reflected light/Incoming light %. Earth's planetary albedo is about 31%. Albedo affects global climate and weather patterns. In 1999, Yogesh Sud, a NASA scientist, and his colleagues found that changes in cloud coverage in the tropics acted as a natural thermostat to keep sea surface temperature (SST) between approximately 28°C and 30°C.

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CLIMATE CHANGE

THE SEVEN (7) THEORIES

5. **Ocean Currents:** due to the slowdown of the ocean's Thermohaline Circulation (THC). Explain global warming through the energy concentration at the upper layer of the oceans, dependent on temperature variations, which give rise to changes in salt content and hence in density.
5. **Planetary motion:** How the Earth's tilt, orbit, and other planets affect the climate, by natural gravitational and magnetic oscillations of the solar system induced by the planet's movement through space. These oscillations modulate solar variations and/or other extraterrestrial influences on Earth, which then drive climate change
5. **Solar Variability:** How changes in the coronal ejections and magnetic fields of the Sun cause changes in cloud formation, ocean currents, wind, etc., and **cause climate change**. Changes in the brightness of the sun are caused by sunspots – bursts of energetic particles and radiation – that vary in frequency in cycles of roughly 11, 87, and 210 years. These cycles cause changes in the amount of electromagnetic radiation – also called "solar wind" – that reaches Earth and its atmosphere, which in turn affects Earth's climate

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PRESENTATION TITLE

IMPACT ON ITS ELEMENTS

- **Water:** Changes in water temperature, water levels, and chemical compositions
- **Soil:** Soil is becoming less fertile and polluted with dangerous chemicals
- **Living Organisms:** While the effects on living organisms are many, the prime concern is that many of them are going extinct
- **Food:** Food is becoming contaminated with hazardous elements, and in some cases, food production is getting lower
- **Human Health:** Various diseases are increasing; ozone layer depletion is increasing the chances of skin cancer





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CLIMATE CHANGE

SO, WHAT DO WE DO?

REDUCE, REUSE, RECYCLE

Production takes a lot of energy. By implementing 3Rs, we can reduce our carbon footprint.

SAVE ENERGY AT HOME

Turn things off when not in use and use energy saving equipment.

CHANGE TRANSPORT METHODS

Use public transport more instead of personal cars. Or just use a bicycle.

PLANT MORE TREES

Trees and forests must be saved at all costs. Plant more trees wherever possible.

EDUCATE PEOPLE AROUND US

Awareness is a must. People should be educated about climate change and its dangers.

AREAS IN WHICH WE SHOULD FOCUS MOST

REDUCING THE PRODUCTION OF GREENHOUSE GASES

Greenhouse gases are one of the **key reasons for climate change**. So, they are one of the first things that should be addressed.

USING RENEWABLE ENERGY SOURCES

Producing energy using fossil fuels creates a lot of problems. They produce GHGs, and there's also the fact that **we will run out of them soon**. The use of renewable energy everywhere should be made a priority.